Year 11 Set 2 · Higher · grade 5–6 · Lesson 1

Simultaneous equations

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Today I will learn to...

Goals

1. Solve linear simultaneous equations by elimination

2. Form and solve simultaneous equations from a problem

Do now: retrieval practice

4 mins

1 Expand $3(2x - 5)$

$6x - 15$

2 Solve $4x + 7 = 31$

$x = 6$

3 Factorise $x^2 + 5x + 6$

$(x + 2)(x + 3)$

4 Make y the subject of $2x + y = 9$

$y = 9 - 2x$

Objective 1

Solve linear simultaneous equations by elimination

Key idea

Learn

!

Make the coefficients of one letter match. Same signs subtract, different signs add. Substitute back, then check in the other equation.

Objective 1: Solve linear simultaneous equations by elimination

I do: example 1

I do

$x + y = 10, x - y = 4$

1. The y terms have opposite signs, so add
2. $(x + y) + (x - y) = 10 + 4 \rightarrow 2x = 14$
3. $x = 7$
4. Substitute: $7 + y = 10$, so $y = 3$

I do: example 2

I do

$3x + 2y = 16, x + 2y = 8$

1. Both have 2y, so subtract
2. $(3x + 2y) - (x + 2y) = 16 - 8 \rightarrow 2x = 8$
3. $x = 4$
4. Substitute: $4 + 2y = 8$, so $2y = 4$, $y = 2$

I do: example 3

I do

$2x + 3y = 13, 3x - y = 3$

1. $\times 3$: $9x - 3y = 9$
2. Add: $11x = 22$, $x = 2$
3. $6 - y = 3$, $y = 3$

Mini whiteboards

We do

1. $2x + y = 11, x + y = 7$

Answer: $x = 4, y = 3$

2. $5x - 2y = 11, 3x + 2y = 13$

Answer: $x = 3, y = 2$

3. $x + y = 10, x - y = 4$

Answer: $x = 7, y = 3$

4. $3x + 2y = 16, x + 2y = 8$

Answer: $x = 4, y = 2$

You do: choose your challenge

You do

Bronze

$x + y = 9$, $x - y = 3$

Silver

$3x + y = 14$, $x + y = 6$

Gold

$5x + 2y = 24$, $2x + 3y = 14$

Platinum

4 pens and 3 rulers cost £3.90. 2 pens and 5 rulers cost £3.70. Find the cost of each.

You do: answers

Check

Bronze

x = 6, y = 3

Silver

x = 4, y = 2

Gold

x = 4, y = 2

Platinum

Pen 60p, ruler 50p

If you finish early

Extra

Bronze

$x + y = 12,\ x - y = 2$

$x = 7,\ y = 5$

Silver

$2x + 3y = 13,\ 2x + y = 7$

$y = 3,\ x = 2$

Gold

$3x + 2y = 19,\ 5x - 4y = 17$

$x = 5,\ y = 2$

Platinum

$4x - 3y = 1,\ 6x + 5y = 49$

$x = 4,\ y = 5$

Exam question 1

4 marks

Solve $4x + 3y = 19$ and $3x - 2y = 10$

Space for working

Mark scheme: ×2 and ×3: 17x = 68, x = 4, y = 1

Exam question 2

4 marks

2 adult and 3 child tickets cost £36. 1 adult and 2 child tickets cost £21. Find the price of each.

Space for working

Mark scheme: 2a + 3c = 36, 2a + 4c = 42 → c = £6, a = £9

Objective 1 achieved

Solve linear simultaneous equations by elimination



Objective 2

Form and solve simultaneous equations from a problem

✓ Achieved: Solve linear simultaneous equations by elimination

Key idea

Learn

!

Choose a letter for each unknown, write two equations from the information, solve by elimination, then answer the question in context.

Objective 2: Form and solve simultaneous equations from a problem

I do: example 1

I do

3 apples and 2 bananas cost £1.30. 1 apple and 2 bananas cost 70p. Find the cost of each.

1. 3a + 2b = 130, a + 2b = 70
2. Subtract: 2a = 60, a = 30
3. 30 + 2b = 70, b = 20
4. Apple 30p, banana 20p

I do: example 2

I do

Two numbers add to 25 and their difference is 7. Find them.

1. $x + y = 25$, $x - y = 7$
2. Add: $2x = 32$, $x = 16$
3. $y = 9$

I do: example 3

I do

2 adult and 1 child ticket cost £26. 1 adult and 3 child tickets cost £28. Find each price.

1. $2a + c = 26$, $a + 3c = 28$
2. $\times 2$: $2a + 6c = 56$
3. Subtract: $5c = 30$, $c = 6$
4. $2a = 20$, $a = 10$. Adult £10, child £6

Mini whiteboards

We do

1. Two numbers sum to 20 and differ by 4. Find them.

Answer: 12 and 8

2. 2 pens + 1 pad = £5. 1 pen + 1 pad = £3.50. Find each.

Answer: Pen £1.50, pad £2

3. Two numbers sum to 15. One is 3 more than the other.

Answer: 6 and 9

4. Adult + child ticket = £14. 2 adults + 1 child = £23.

Answer: Adult £9, child £5

You do: choose your challenge

You do

Bronze

Two numbers add to 30 and differ by 8. Find them.

Silver

3 cups and 1 plate cost £11. 1 cup and 1 plate cost £5. Find each price.

Gold

A farm has cows and chickens: 20 heads and 56 legs. How many of each?

Platinum

The line $y = mx + c$ passes through (1, 5) and (3, 11). Find m and c.

You do: answers

Check

Bronze

19 and 11

Silver

Cup £3, plate £2

Gold

$c + h = 20$, $4c + 2h = 56$. 8 cows, 12 chickens

Platinum

$m + c = 5$, $3m + c = 11$. $m = 3$, $c = 2$

If you finish early

Extra

Bronze

Two numbers add to 25 and differ by 7. Find them.

16 and 9

Silver

5 apples and 2 pears cost £1.60. 3 apples and 2 pears cost £1.20. Find the cost of each.

Apple 20p, pear 30p

Gold

A test has 20 questions: +3 for right, −1 for wrong. Mia answers all of them and scores 40. How many did she get right?

r + w = 20, 3r − w = 40, so r = 15

Platinum

The line y = mx + c passes through (2, 7) and (5, 16). Find m and c.

2m + c = 7, 5m + c = 16: m = 3, c = 1

Exam question 1

4 marks

5 coffees and 2 teas cost £14.60. 3 coffees and 2 teas cost £9.80. Find the cost of one coffee and one tea.

Space for working

Mark scheme: 2c = 4.80, coffee £2.40, tea £1.30

Objective 2 achieved

Form and solve simultaneous equations from a problem



Real life

Apply

3 coffees and 2 cakes cost £13.40. 1 coffee and 4 cakes cost £11.80. Find the price of each.

Coffee £3.00, cake £2.20

Extension: stretch yourself

Stretch

1. Show that $y = x + 5$ and $y = x^2 + x + 6$ do not meet.

$x^2 + 1 = 0$ has no real solutions

2. Solve $x^2 + y^2 = 25$ and $y = x + 1$

$x^2 + x - 12 = 0 \rightarrow (3, 4)$ and $(-4, -3)$

Exit ticket: exam questions

Exit

1. Solve $2x + y = 10$ and $x - y = 2$ **(3 marks)**

2. Solve $3x + 4y = 5$ and $2x - 3y = 9$ **(4 marks)**

3. 4 notebooks and 3 pens cost £7.10. 2 notebooks and 3 pens cost £4.90. Find the cost of a notebook and a pen. **(4 marks)**

Exit ticket: mark scheme

Mark it

Q1 x = 4, y = 2 [3]

Q2 17x = 51, x = 3, y = −1 [4]

Q3 2n = 2.20, notebook £1.10, pen 90p [4]

Total: ____ / 11

Today I achieved

- ✔ **Solve linear simultaneous equations by elimination**
- ✔ **Form and solve simultaneous equations from a problem**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Solve $x + 3y = 11$ and $x + y = 5$ [3]

2. Solve $4x - y = 10$ and $2x + y = 8$ [3]

3. Solve $2x + 5y = 16$ and $3x - 2y = 5$ [4]

4. 3 adult and 2 child tickets cost £44. 2 adult and 3 child tickets cost £41. Find the price of each ticket. [4]

5. 2 teas and 3 cakes cost £9.50. 4 teas and 1 cake cost £8.50. Find the cost of each. [4]

Extra practice: Maths Genie → Edexcel question bank → Simultaneous Equations